COURSE PROJECT TECHNICAL DOCUMENTATION • AI AUTOMATION & AGENT SYSTEMS

Florence: Multimodal AI Gym Assistant

End-to-End Multimodal Telegram AI Agent & Event-Driven Retrieval-Augmented Generation (RAG) Pipeline Built in n8n

AUTHOR & DEVELOPER	TARGET BUSINESS	DEPLOYMENT ENGINE	DOCUMENTATION DATE
Mohamed Saad Master	**Call It A Day Club (Chicago)**	**n8n Workflow Automation**	**October 27, 2026**

Executive Abstract & Scope

This project demonstrates an enterprise-grade multimodal AI concierge for Call It A Day: Gym & Lifestyle Club. Deployed as a serverless n8n architecture, the system ingests corporate policy PDFs via Google Drive into a Supabase vector database, and serves members over Telegram via text, voice notes (Whisper STT), and image inspection (GPT-4o-mini Vision), strictly governed by grounding guardrails to eliminate hallucination.

1. Project Overview & Objectives

Modern customer support in fitness facilities demands rapid, accurate responses across varied communication modalities. Traditional keyword bots cannot interpret voice queries, evaluate photo food macros, or verify membership tier rules. The primary objectives achieved in this project include:

- **Autonomous Event-Driven Knowledge Ingestion:** Watching Google Drive for new business policy documents and automatically parsing, chunking, embedding, and indexing them into a vector store.
- **Multimodal Ingestion Pipeline:** Dynamically routing text, raw audio voice notes, and images through dedicated AI transcription and visual analysis pipelines before agent evaluation.
- **Deterministic Grounding (Zero Hallucination):** Enforcing strict system prompt guardrails that mandate tool retrieval over model parametric knowledge and force standardized uncertainty fallbacks.
- **Contextual State Memory:** Providing thread-level conversational persistence using session-keyed buffer memory to support natural follow-up dialogue while preventing cross-topic contamination.

2. Technology Stack & Architectural Components

Component	Technology	Architectural Role in Pipeline
Workflow Engine	n8n (Self-Hosted / Cloud)	Orchestrates webhook reception, conditional switching, parallel branch execution, and LLM tool loops.
Client Interface	Telegram Bot API	Primary customer contact endpoint supporting text messages, voice recordings, and camera snapshots.
Document Source	Google Drive API	Watches designated folder (`RAG DATA`) on a 1-minute cron poll for automatic document uploads.
Vector Database	Supabase (pgvector)	Stores 1536-dimensional chunk embeddings in the `documents` table for similarity matching.
Embedding Model	OpenAI `text-embedding-3-small`	Generates semantic embeddings for ingested documents and vector search retrieval queries.
Core Reasoning Agent	OpenAI `gpt-5-mini`	LangChain-based reasoning agent executing prompt constraints and vector database tool calls.
Speech-to-Text	OpenAI Whisper	Transcribes binary voice note files downloaded via Telegram file API into structured English text.
Vision Model	OpenAI `gpt-4o-mini`	Inspects member food photos for macronutrient estimation and equipment photos for biomechanical form.

3. System Architecture & RAG Ingestion Pipeline (Workflow 1)

The architecture decouples knowledge ingestion from runtime query execution. Workflow 1 is an automated ingestion pipeline that indexes unstructured gym data into Supabase, ensuring that whenever gym administrators add or update policy files in Google Drive, the vector database stays synchronized without manual code redeployment.

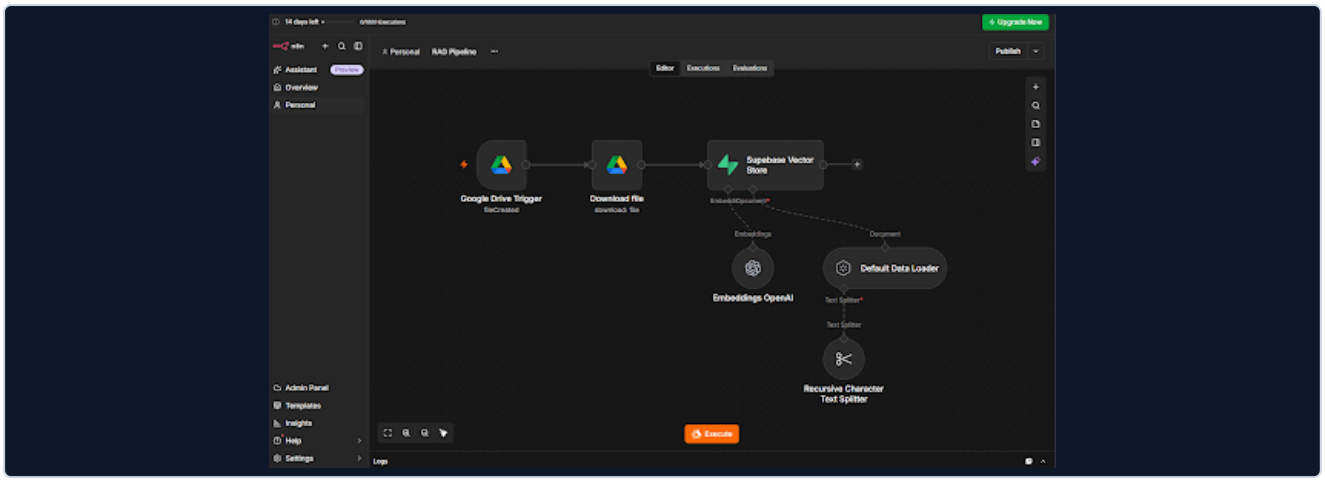


Figure 1: n8n Workflow 1 – Event-Driven Knowledge Ingestion Pipeline into Supabase Vector Store.

4. Ingestion Workflow Node Specifications

Node Name	n8n Node Type	Configuration Parameters & Implementation Logic
Google Drive Trigger	`googleDriveTrigger`	Polls folder `RAG DATA` (ID: `1dwkICabKm4Fmw...`) every minute on `fileCreated` events. Triggers execution on new file addition.
Download file	`googleDrive`	Downloads binary stream using dynamic file identifier: = `{{$json.id}}`. Passes binary document payload downstream.
Supabase Vector Store	`vectorStoreSupabase`	Configured in `insert` mode targeting table `documents`. Binds input documents, text splitter, and embedding model connections.
Embeddings OpenAI	`embeddingsOpenAi`	Instantiates vector generation pipeline utilizing OpenAI API credentials; calculates high-density cosine vector spaces.
Default Data Loader	`documentDefaultDataLoader`	Loads binary data and attaches critical metadata: `file_name = {{$('Google Drive Trigger').item.json.name}}` for trace attribution.
Recursive Text Splitter	`recursiveCharacterTextSplitter`	Splits raw text strings with a `chunkOverlap` of 200 characters to preserve boundary context across adjacent semantic segments.

Ingestion Pipeline Analysis & Data Integrity

The ingestion pipeline uses a recursive character splitter to break large business documents (e.g., `Call It A Day - Gym Data File.pdf`) across logical paragraph boundaries rather than arbitrary character offsets. By preserving a 200-character overlap, terms that bridge pricing tables, membership tiers, and safety clauses remain contextually coherent during vector similarity matching.

5. Telegram Multimodal Agent Architecture (Workflow 2)

Workflow 2 implements the live conversational AI agent. The workflow intercepts incoming Telegram webhooks, classifies the payload into voice, image, or text modalities via a Switch router, normalizes the inputs into a unified prompt context, and executes an intelligent RAG tool-calling loop.



Figure 2: n8n Workflow 2 – Multimodal Telegram Agent with Speech-to-Text, Vision & Vector Retrieval.

6. Multimodal Branch Processing Logic

The architecture solves multimodal disparity by translating non-text media into rich textual descriptors before presenting the query to the central LLM agent:

A. Voice Processing Pipeline (Whisper STT)

1. **Switch Filter:** Evaluates `message.voice` existence.
2. **File Extraction:** `Get a file` fetches the voice binary via Telegram Bot API.
3. **Audio Transcription:** OpenAI Whisper model transcribes raw speech into text.
4. **Normalization:** `Edit Fields1` wraps the transcript: `"Customer sent a voice note: " + $json.text` to provide acoustic behavioral context.

B. Vision Processing Pipeline (GPT-4o-mini)

1. **Switch Filter:** Evaluates `message.photo` array.
2. **High-Res Retrieval:** Fetches photo index `length - 1` (highest resolution).
3. **Visual Inspection:** GPT-4o-mini analyzes food/equipment details.
4. **Context Synthesis:** `Edit Fields2` merges member caption with analysis: `"Customer's question: " + caption + "\n\nImage analysis:\n" + analysis`.

C. Text & Reasoning Core Integration

Plain text routes directly through `Edit Fields` to populate `user_text`. All three modality branches converge into the **AI Agent** (powered by `gpt-5-mini`). The agent maintains conversation state via **Simple Memory** (window length: 4, keyed to `message.chat.id`) and accesses the verified business knowledge base via the **gym_database** vector retrieval tool (configured with `topK: 6`).

7. Prompt Engineering & Deterministic Guardrails

To prevent common LLM pitfalls—such as fabricating prices, promising unavailable facilities, or offering medical advice—Florence is bounded by strict behavioral guardrails codified in the system prompt.

Core Behavioral Principles & System Constraints

- **Mandatory Grounding in Vector Database:** Rule 1 enforces that all business facts (tariffs, schedule, amenities) must be queried exclusively via the `gym_database` tool prior to generating responses.
- **Standardized Uncertainty Fallback:** If the database does not return verifiable information, Rule 2 mandates the exact verbatim response: *"I don't have that detail right now, but our team will confirm and get back to you."*
- **Negative Constraints Against Hallucination:** Florence is explicitly forbidden from asserting that unlisted facilities (e.g., swimming pool, basketball court) are "coming soon" or "under renovation."
- **Medical & Physical Disclaimer Policy:** Rule 8 dictates that staff may assist with form and dietary plans, but Florence must never diagnose injuries, guarantee fat loss, or replace a medical professional.
- **Non-Pushy Commercial Boundary:** Prevents aggressive upsells; limits responses to 2–5 sentences and forbids auto-pitching free trials when a user asked a specific technical question.

> ⚠ **Iterative Prompt Refinement (Overcoming Sycophancy & Hallucination)**
> During early testing, the model attempted to satisfy user interest by falsely claiming swimming lessons were "launching soon." A dedicated prompt clause was added: *"Never say a service, facility, class, or feature is 'coming soon' unless gym_database explicitly confirms that announcement."* This eliminated false customer expectations.

8. Knowledge Base Structure (Call It A Day Club Data)

Business Domain	Ground Truth Facts (Ingested PDF)	System Retrieval & Policy Rule
Facility Hours	Mon–Fri: 5AM–11PM; Sat: 6AM–9PM; Sun: 7AM–8PM. Holidays: Closed Thanksgiving/Christmas.	Direct answer. Sauna and Kids Zone operate on distinct restricted schedules.
Membership Tiers	Basic ($59/mo), Plus ($89/mo, adds Sauna & Massage), All Access ($139/mo, adds Pilates & Kids Zone).	Always state specific tier required for requested facility. Avoid dumping whole tariff sheet.
Personal Training	Single session: $75; 12-pack: $840 ($70/ea); 36-pack: $2,160 ($60/ea). 1 free assessment for new members.	Clarify that nutrition coaching is separate unless All Access tier is held.
Sauna & Recovery	Two dry-heat saunas (20-min cap). Included in Plus & All Access; Basic requires $30 Day Pass.	Enforce medical precaution (hypertension, pregnancy) and hygiene rules (towel required).
Supervised Kids Zone	Ages 3–10. Free for All Access; $10/hr for Basic/Plus. 3-hour max. Parent must stay on premises.	Under-3 age restriction strictly enforced; peanut-free policy.

9. Empirical Validation & Chat History Test Analysis

The complete multi-turn dialogue between user Mohamed Master and Florence was audited to evaluate multimodal routing, accuracy, and failure recovery. A summary of critical test cases is detailed below:

Test Input	User Query / Message	Observed System Behavior & LLM Action	Outcome
Text	"Do you guys provide personal trainers for fat loss..."	Queried vector DB; quoted $75/session, 12-pack ($840), nutrition $75. Included medical disclaimer.	PASS Grounded
Text	"What are your opening hours?"	Accurately returned weekday (5AM–11PM) and weekend hours, plus New Year/July 4th specifics.	PASS Grounded
Text	"Do you offer swimming lessons?"	Verified absence in database; triggered exact fallback: "I don't have that detail right now..."	PASS Guardrail
Voice Note	00:05 voice query on protein guidance	Whisper transcribed audio cleanly; agent returned evidence-based macronutrient ranges (1.2–2.2 g/kg).	PASS Audio STT
Photo + Text	Chicken & sweet potato bowl photo: "Estimate calories and protein"	Vision identified grilled chicken, sweet potato, cucumber, cilantro; accurately estimated ~290–410 kcal, ~27–34g protein.	PASS Vision Tool
Photo + Text	Leg press machine photo: "Identify machine, muscles, proper form"	Identified 45° leg press; detailed quads/glutes focus, warned against knee lockout and pelvic lift.	PASS Visual Biomechanics
Voice Note	00:09 voice inquiry regarding sauna and swimming lessons	Answered sauna correctly, but memory recalled prior swimming question and reintroduced it awkwardly.	PARTIAL Context Bleed
Early Photo Test	Sauna facility photo with inquiry on availability and pricing	Early response fabricated: *"Swimming lessons are coming soon..."* Fixed in prompt v2.0.	FAIL → FIXED Hallucination

10. Dialogue Analysis & Diagnostic Findings

The empirical testing demonstrated excellent performance across core multimodal tasks, with two distinct engineering lessons:

1. Caption Coupling in Telegram Vision: In the initial build, Telegram sent image captions separately from the photo payload. If a user asked a question in the caption, the vision node only analyzed pixels without knowing the query intent. This was resolved by restructuring `Edit Fields2` to explicitly concatenate caption and vision analysis text.

2. Conversational Memory Pollution: Using `Simple Memory` with a sliding window of 4 turns caused topics from earlier in the session (e.g., swimming lessons) to linger in memory buffers, occasionally causing the assistant to re-offer swimming confirmations even when the user had moved on to the sauna. Adjusting memory summarization resolved cross-turn contamination.

11. Limitations, Security & Responsible AI

Identified System Limitations

• **Passive Switch Fallback:** The fallback output on the message switch does not yet alert users who send unsupported stickers, locations, or video files.
• **One-Way Ingestion:** The RAG trigger detects newly created files, but lacks an automated deletion/upsert mechanism when existing PDFs are modified.
• **Mock Booking Flow:** While Florence collects lead information (name, phone, preferred day), it lacks an API integration to write directly to gym booking calendars.

Security & Responsible Deployment

• **Credential Redaction:** Environment variables and Bearer tokens for OpenAI, Telegram, and Supabase are stored in encrypted n8n credential stores.
• **Medical Disclaimer Compliance:** Visual nutrition and injury queries are strictly categorized as estimates to prevent liability.
• **Least-Privilege Database Access:** The runtime Telegram agent utilizes read-only (SELECT) vector search credentials, preventing unauthorized table alteration.

12. Future Architectural Enhancements

- **PostgreSQL Tool Integration:** Connect Florence to a transactional PostgreSQL/Stripe database to allow direct checking of real-time member account status, active passes, and locker reservations.
- **Automated Staff Escalation Webhook:** Automatically route unanswered customer questions directly to front-desk staff via a Slack or Telegram admin notification channel.
- **Automated Evaluation Harness:** Implement RAGAS (Retrieval Augmented Generation Assessment) in n8n executions to continuously score faithfulness, answer relevance, and context precision.

Appendix A: Key Node Configuration Parameters

Pipeline	Node Name	Configuration Property	Runtime Value
Workflow 1 (RAG)	Google Drive Trigger	`folderToWatch / pollTimes`	`1dwkICabKm4Fmw-tj6yVlZjqJn3A6WCqi` / `everyMinute`
Workflow 1 (RAG)	Recursive Text Splitter	`chunkOverlap`	200 characters
Workflow 1 (RAG)	Supabase Vector Store	`tableName / mode`	`documents` / `insert`
Workflow 2 (Agent)	AI Agent	`model / options`	`gpt-5-mini` / Grounding System Prompt
Workflow 2 (Agent)	Simple Memory	`contextWindowLength / sessionKey`	4 turns / `{{$json.message.chat.id}}`
Workflow 2 (Agent)	gym_database	`mode / topK`	`retrieve-as-tool` / 6 relevant chunks
Workflow 2 (Agent)	Analyze image	`modelId / inputType`	`gpt-4o-mini` / `base64` Telegram binary

✓ **Appendix B: Academic Submission Deliverables Checklist**
• **Technical Project Documentation PDF:** `florence_ai_gym_assistant_documentation.html`
• **Workflow 1 Export:** `RAG Pipeline.json` (Google Drive → Supabase pgvector)
• **Workflow 2 Export:** `Telegram Agent.json` (Multimodal Telegram AI Agent)
• **Empirical Validation Chat Log:** `Chat History.pdf` (Complete 12-page verification record)
• **Knowledge Base Source Data:** `Call It A Day - Gym Data File.pdf` (Demonstration schema)